

The Debt-Inflation Channel of the German Hyperinflation

Brunnermeier, Correia, Luck, Verner & Zimmermann (2023, *WP*)

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Motivation

- How does **inflation** transmit to the **real economy**?
- What is the role of a **debt-inflation channel** through financing frictions?
 - Inflation
 - wealth redistribution to net nominal debtors
 - real effects by relaxing financing constraints: Keynes (1923), Fisher (1933)
- How important is this financial channel relative to the **NK channel** through price/wage rigidity?
- This Paper:
 - Use the German inflation of 1919-1923 to study how a “big” inflation transmits to the real economy through the debt-inflation channel

The German Inflation, 1919-1923

- Canonical Event in Monetary History
 - Depreciation: 4.2 marks/dollar → 4.2 *trillion* marks/dollar
 - Studied by generations of economists to understand the fundamental causes of inflations + macroeconomic and distributional effects
Keynes (1923), Graham (1930), Bresciani-Turroni (1937), Cagan (1956), Sargent (1982), Kindleberger (1985), Dornbusch and Fischer (1986), Feldman (1993)...
- Features of the empirical setting
 1. Inflation was **unanticipated**
 2. Massive inflation puts **debt-inflation channel into sharp relief**
 3. Newly digitized micro data → **disentangle mechanisms**

Key Findings

1. Debt-inflation channel: aggregate evidence

- Inflation led to a large **fall in leverage** (50%) and interest expense share (60%)
- Large **decline in bankruptcies** (70%)

2. Debt-inflation channel: firm-level evidence

- **Redistribution**: High-leverage firms see relative increase in book and market equity values, along with fall in interest expenses
- **Real effects**: High leverage firms boost employment (75% of overall increase)

3. Nominal rigidities channel

- **Frequency of wage and price adjustment increasing** with inflation
- Debt-inflation channel is active even with flexible prices

Structure of the Presentation

Conceptual Framework

Historical & Macroeconomic Background

Firm-Level Evidence on the Debt-Inflation Channel

Conclusion

Conceptual Framework

- We consider a static model with two subperiods: “morning” and “evening”
- The economy is populated by
 - a unit mass of risk neutral entrepreneurs, who operate the productive technology
 - workers, who monopolistically provide labor to firms, $L_i = \left(\int_0^1 (L_{ij})^{\frac{\varepsilon-1}{\varepsilon}} dj \right)^{\frac{\varepsilon}{\varepsilon-1}}$

Firm's Problem - Default Decision

- Given initial K_0, D_0 and $Z_i \sim G$, the entrepreneur solves

$$\max \left\{ K_0 - Z_i - \frac{D_0}{P} + V, 0 \right\}$$

where V is the value of operating the firm and P is the price level

- The cutoff value for default is given by

$$Z^* = K_0 - \frac{D_0}{P} + V(Z^*)$$

which implies that there is a $G(Z^*)$ measure of active entrepreneurs

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Firm's Problem - Input and Financing Choices

- Each firm i with $Z \leq Z^*$ solves

$$\begin{aligned} P \cdot V &= \max_{K_i, L_i, D_i} PAK_i^\alpha L_i^{1-\alpha} - WL_i - PK_i \\ \text{s.t. } [FF] \quad &D_i - D_0 - Z_i = WL_i + P(K_i - K_0) \\ [WC] \quad &D_i \leq \xi PAK_i^\alpha L_i^{1-\alpha} \end{aligned}$$

- There are two solutions to the problem, depending on whether the working capital constraint binds or not. Assuming a parametrization of G and ξ such that all firms are constrained, we can rewrite the problem as

$$\begin{aligned} P \cdot V &= \max_{K_i, L_i, D_i} PAK_i^\alpha L_i^{1-\alpha} - WL_i - PK_i \\ \text{s.t.} \quad &WL_i + P(K_i - K_0) + D_0 + Z_i = \xi PAK_i^\alpha L_i^{1-\alpha} \end{aligned}$$

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Firm's Problem - Solution

- Then, we have

$$F_K - 1 = \lambda [1 - \xi F_K]$$

$$F_L - \frac{W}{P} = \lambda \left[\frac{W}{P} - \xi F_L \right]$$

which implies

$$\frac{K_i}{L_i} = \frac{\alpha}{1 - \alpha} \frac{W}{P}$$

- Labor demand is then given by

$$L_i^d = \frac{K_0 - \frac{D_0}{P} - \frac{Z_i}{P}}{\frac{1}{1-\alpha} \frac{W}{P} - \xi A \left(\frac{\alpha}{1-\alpha} \frac{W}{P} \right)^\alpha} \implies L^d = G(Z^*) \frac{K_0 - \frac{D_0}{P} - \frac{1}{P} \frac{\int Z^{Z^*} Z dG(Z)}{G(Z^*)}}{\frac{1}{1-\alpha} \frac{W}{P} - \xi A \left(\frac{\alpha}{1-\alpha} \frac{W}{P} \right)^\alpha}$$

- The real value of production is

$$V = \frac{D_0}{P} + Z_i - K_0 + (1 - \xi) A K_i^\alpha L_i^{1-\alpha}$$

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Result 1 – $P \uparrow \implies \text{Defaults} \downarrow$

Given the real value of production, the value of not defaulting is

$$J = (1 - \xi) A K_i^\alpha L_i^{1-\alpha} = (1 - \xi) A \left(\frac{\alpha}{1 - \alpha} \right)^\alpha \left(\frac{W}{P} \right)^\alpha L_i^d$$

As such, an entrepreneur defaults if they would like to choose $L_i^d \leq 0$ and, thus, $Z^* = K_0 - \frac{D_0}{P}$ and the amount of defaults is given by $1 - G(Z^*)$

Result 1. The share of defaulting firms declines for higher levels of inflation, P :

$$\frac{\partial (\text{Default Share})}{\partial P} = -\frac{D_0}{P^2} G' \left(K_0 - \frac{D_0}{P} \right) < 0$$

Workers Problem

- The household – the collection of all workers $j \in [0, 1]$ – solves

$$\begin{aligned} \max_{C, L} \quad & \ln(C) - \chi \frac{L^{1+\varphi}}{1+\varphi} \\ \text{s.t.} \quad & C = \frac{W}{P} L + \frac{G(Z^*) D_0}{P} \end{aligned}$$

- Each worker sets W_j at which they are willing to work. Given the CES structure and L^d , we have that $L(W_j) = \left(\frac{W_j}{W}\right)^{-\varepsilon} L^d$ which implies that

$$\frac{W}{P} = \frac{\varepsilon}{\varepsilon - 1} \chi (L)^\varphi C$$

- Aggregating workers' choices, we have that L^s satisfies

$$\frac{W}{P} = \frac{\frac{\varepsilon}{\varepsilon - 1} \chi (L^s)^{1+\varphi}}{1 - \frac{\varepsilon}{\varepsilon - 1} \chi (L^s)^{1+\varphi}} \left(\frac{G(Z^*) D_0}{P} \right)$$

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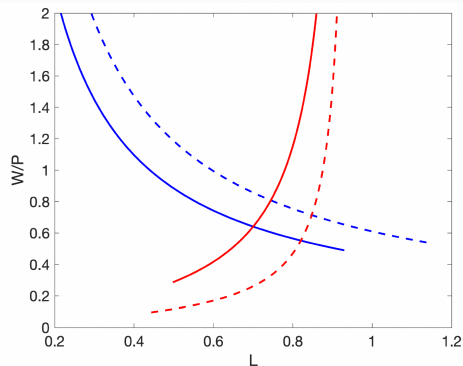
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Flexible Wage Equilibrium

- With flexible wages, the equilibrium in the labor market is given by L^* that satisfies the L^d and L^s equations simultaneously
- The aggregate resource constraint is

$$\int_{\underline{Z}}^{Z^*} AK_i^\alpha L_i^{1-\alpha} dG(Z_i) = \int_{\underline{Z}}^{Z^*} [C_{ie} + K_i - K_0] dG(Z_i) + C_w$$

low (solid) and high (dashed) of P



Result 2: Inflation boosts L^d if firms have nominal debt are constrained

We have

$$L^d = G(Z^*) \frac{K_0 - \frac{D_0}{P} - \frac{\int_{\underline{Z}}^{Z^*} Z dG(Z)}{G(Z^*)}}{\frac{1}{1-\alpha} \frac{W}{P} - \xi A \left(\frac{\alpha}{1-\alpha} \frac{W}{P} \right)^\alpha}$$

So, $P \uparrow \implies$ employment $\uparrow$ and output $\uparrow$

The higher the level of initial debt D_0 , the stronger the increase in labor demand and, thereby, the debt-inflation channel.

The increase in the real wage is offset by the wealth effect on labor supply from the erosion of workers' real debt holdings.

Result 3: As $P \uparrow$, Debt-Inflation becomes more relevant

Introduce nominally rigidity by assuming that initially, the wage for all workers is set at W_0 , which we assume is the equilibrium flexible wage with $P = 1$.

To alter W_j , need to incur a cost $\psi \geq 0$.

This will generate different behavior of the economy in times of normal inflation vs hyperinflation:

- For a small increases, no update: $W/P \downarrow$, and employment $\uparrow\uparrow\uparrow$ through both the nominal rigidity and debt-inflation channels.
- For a large increases, the wage is updated: inflation only affects real outcomes through the debt-inflation channel.

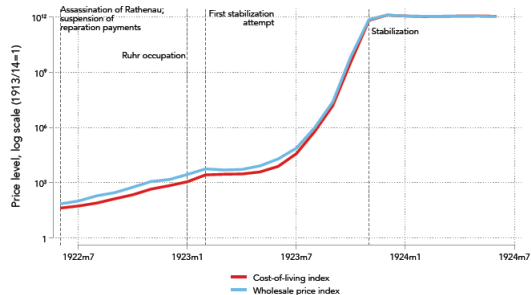
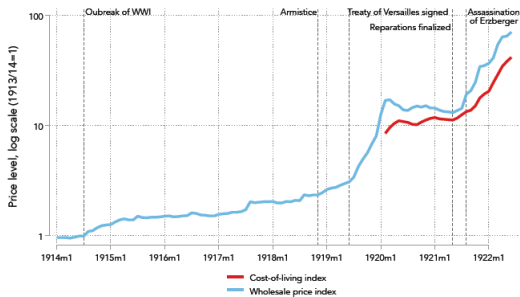
1. **Unexpected inflation** $\implies$ **decline in bankruptcies**
2. **Debt-inflation channel:** Relaxing financing constraint $\implies$ increase in output
 - Especially for firms with high leverage and/or with long-term debt (Kiyotaki and Moore 1997, Cordoba and Ripoll 2004, Gomes, Jermann, and Schmid 2016)
3. **Nominal rigidity channel:**
 - Low inflation: boosts output by reducing real wage
 - High inflation: wages become flexible, and inflation only has real effects through the financial channel

Historical & Macroeconomic Background

Background on Weimar Germany's Inflation

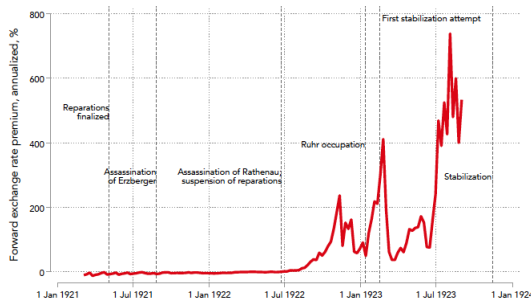
- Roots of inflation: WWI
 - Gold standard abandoned in August 1914
 - Deficit financed war spending
- Two phases of the **post-war inflation (1919-1923)**:
 1. High inflation: WWI Armistice to June 1922
 2. Hyperinflation: July 1922 to November 1923
- Key factors:
 - Fiscal: Reparations, large deficits: "Tanzi effect"
 - Political economy: Lack of political will to raise taxes and cut spending
 - Monetary: Reichsbank accommodated spending by discounting government securities
 - Balance of payments: FX depreciation fueled inflation (departure from PPP)

The Price Level in Germany



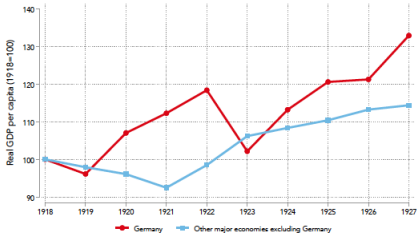
Inflation Expectations

- Average inflation during 1870-1914: 0.7%
- Before June 1922: Expectations of mark appreciation (Kindleberger 1985)
 - Speculation on the mark
 - Bank credit available
- After June 1922: Expectations of mark depreciation
 - Flight from the mark
 - Credit supply dries up

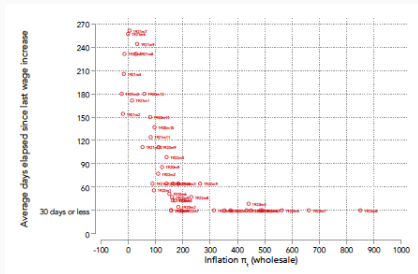


Booming Economy until the End of 1922

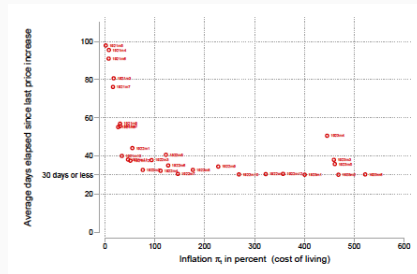
- Strong growth and low unemployment until the end of 1922
 - Germany avoids 1920-21 "Depression"
- Contraction in 1923: hyperinflation, Ruhr crisis, and stabilization



Increasing Frequency of Wage and Price Adjustment



Declining interval between wage increases



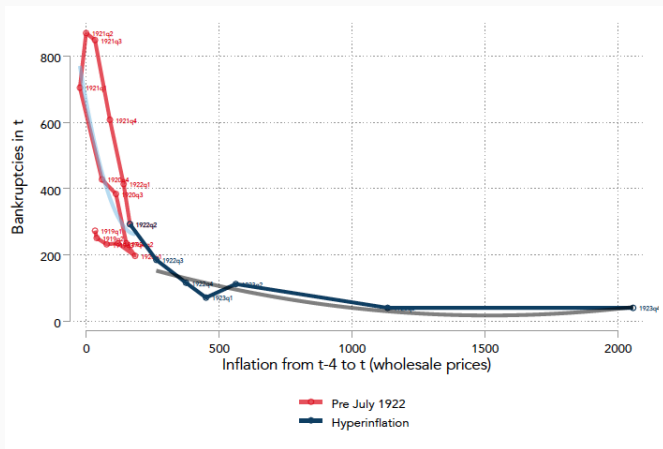
Declining interval between price increases

- Consistent with menu cost models (Alvarez et al 2019)
- Implication: Limited scope for expansionary effect through nominal rigidity channel (Goloso and Lucas 2007)

Firm-Level Evidence on the Debt-Inflation Channel

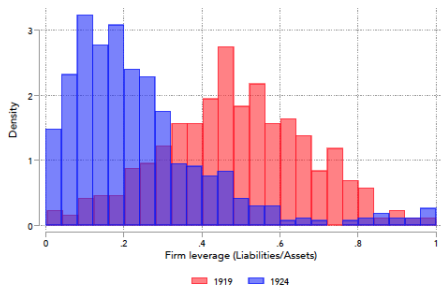
Inflation and Bankruptcies

- Reduction in leverage and interest costs reduced likelihood of financial distress,

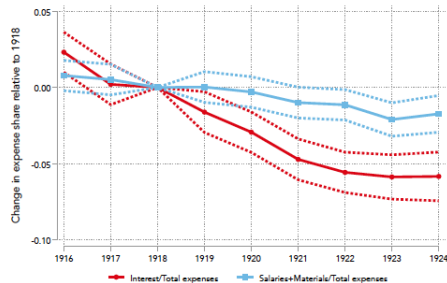


Hyperinflation Led to Collapse in Leverage and Interest Expenses.

(a) Distribution of leverage in 1919 and 1924.



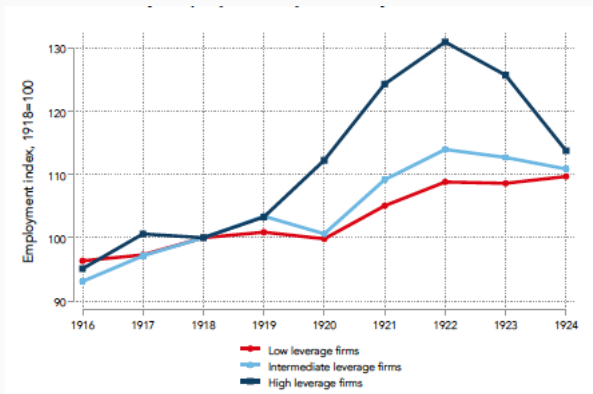
(b) Evolution of expense shares over time.



Firm-Level Analysis

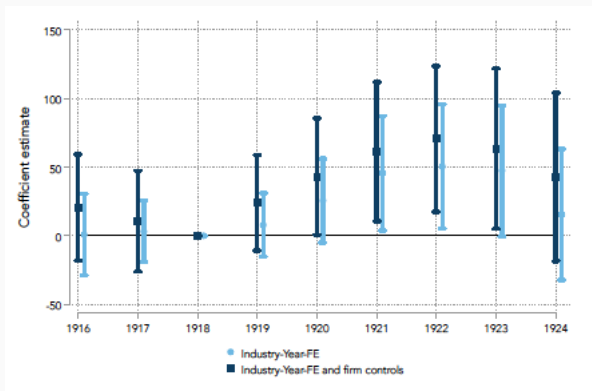
- Is the debt-inflation channel present in the cross-section of firms?
 - Does debt-inflation merely redistribute wealth or does it affect real activity?
- Firm-level measures of exposure to the debt-inflation channel, as of 1918-19
 - $Liabilities/Assets = 1 - \frac{BookEquity}{Assets}$
 - $Debt/Assets$
- Advantages of firm-level analysis:
 - Identification: Control for time-varying industry-specific shocks + firm characteristics
 - Quantify extent of redistribution
 - Estimate real effect on employment

Employment Dynamics across High and Low Leverage Firms



Employment Dynamics across High and Low Leverage Firms

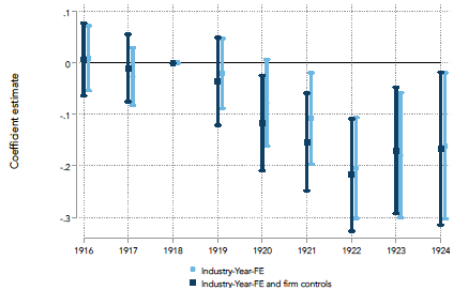
$$\ln(\text{Employment}_{it}) = \alpha_i + \gamma_{st} + \sum_{k \neq 1918} \beta_k \left(\frac{\text{Liabilities}}{\text{Assets}} \right) \mathbf{1}_{k=t} + \sum_{k \neq 1918} X_i \Gamma_k \mathbf{1}_{k=t} + \varepsilon_{it}$$



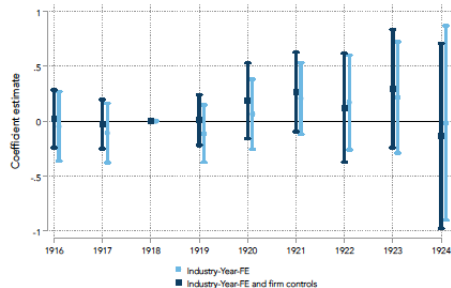
Firm Leverage, Interest Expenses, and Material Expenses and Salaries

$$y_{it} = \alpha_i + \gamma_{st} + \sum_{k \neq 1918} \beta_k \left(\frac{\text{Liabilities}}{\text{Assets}} \right) \mathbf{1}_{k=t} + \sum_{k \neq 1918} X_i \Gamma_k \mathbf{1}_{k=t} + \varepsilon_{it}$$

(a) Interest Expenses.



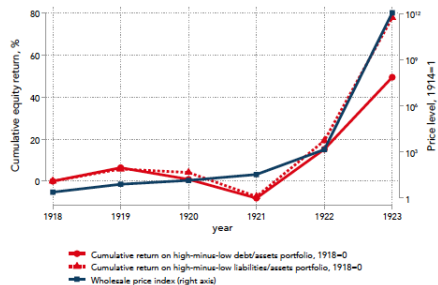
(b) Materials and Salaries.



High Leverage Firms' Stock Returns Outperformed Low Leverage Firms during the Inflation

$$Return_{it} = \gamma_t + \beta Leverage_{it-1} + \varepsilon_{it}$$

	(1)	(2)	(3)	(4)	(5)	(6)
Liabilities/Assets _{<i>i,t-1</i>}	19.0*** (5.47)	16.7*** (5.89)	19.7*** (6.50)			
Debt/Assets _{<i>i,t-1</i>}				16.1*** (5.10)	14.5*** (5.25)	16.5*** (5.48)
Observations	2739	2735	2735	2740	2736	2736
R ²	0.71	0.76	0.76	0.71	0.76	0.76
Year Fixed Effects	✓			✓		
Industry-Year FE		✓	✓		✓	✓
Firm Controls			✓			✓



Stronger Effects for Firms with More Long-Term Debt

$$y_{it} = \alpha_i + \delta_{st} + \sum_{s \neq 1} \beta_{1,s} \mathbf{1} \left[\left(\frac{LTD}{Debt} \right)_{i,1918-1919} \in Q_s \right] \times \left(\frac{Debt}{Assets} \right)_{i,1918-1919} \times \mathbf{1}_{t \geq 1920} \\ + \beta_2 \times \left(\frac{Debt}{Assets} \right)_{i,1918-1919} \times \mathbf{1}_{t \geq 1920} + \Gamma \times (X_i \times \mathbf{1}_{t \geq 1920}) + \varepsilon_{it}$$

Dependent Variable	Int. Expenses/Tot. Expenses			ln(Employment)		
	(1)	(2)	(3)	(4)	(5)	(6)
Debt/Assets _{i,1918-1919} × $\mathbf{1}_{t \geq 1920}$	0.019 (0.049)	0.019 (0.049)	0.012 (0.055)	16.0 (26.4)	16.0 (26.4)	11.4 (30.6)
Q ₂ × Debt/Assets _{i,1918-1919} × $\mathbf{1}_{t \geq 1920}$	-0.096 (0.079)	-0.10 (0.078)	-0.084 (0.078)	21.3 (51.4)	19.9 (51.8)	12.2 (50.0)
Q ₃ × Debt/Assets _{i,1918-1919} × $\mathbf{1}_{t \geq 1920}$	-0.19** (0.083)	-0.19** (0.083)	-0.13 (0.083)	25.5 (32.8)	25.5 (32.8)	26.1 (35.3)
Q ₄ × Debt/Assets _{i,1918-1919} × $\mathbf{1}_{t \geq 1920}$	-0.32*** (0.090)	-0.33*** (0.088)	-0.31*** (0.089)	64.9* (33.9)	64.9* (33.9)	70.4* (40.6)
Observations	3482	3480	3482	1938	1937	1937
Number of Firms	547	547	547	360	360	360
R ²	0.67	0.67	0.72	0.97	0.97	0.97
Year FE	✓			✓		
Firm FE	✓	✓	✓	✓	✓	✓
Industry-Year-FE	✓	✓	✓	✓	✓	✓
Controls × $\mathbf{1}_{t \geq 1920}$			✓			✓

Conclusion

Conclusion

- Aggregate and firm-level evidence of debt-inflation from the German inflation
 - Redistribution toward net debtor firms → real effects
 - Key frictions: long-term nominal debt and financing constraints
- Aggregate effects?
 - PE aggregation: 14% employment increase (75% of total) → debt-inflation is expansionary even with flexible prices
 - Reduction in credit supply → next paper
- External validity?
 - Relevant channels may differ during smaller inflations
 - Monetary policy response
 - Structure of debt contracts matters (fixed/floating, long-term/short-term, domestic/FX)

Appendix

Result 3: As $P \uparrow$, Debt-Inflation becomes more relevant

Introduce nom. rigidity by assuming that initially W_0 , the equilibrium flexible wage with $P = 1$.

To alter W_j , need to incur a cost $\psi \geq 0$.

This has different implications in times of normal inflation vs hyperinflation:

- For a small increases, no update: $W/P \downarrow$, and employment $\uparrow\uparrow\uparrow$ through both the nominal rigidity and debt-inflation channels.
- For a large increases, the wage is updated: inflation only affects real outcomes through the debt-inflation channel.

Green points denote equilibria

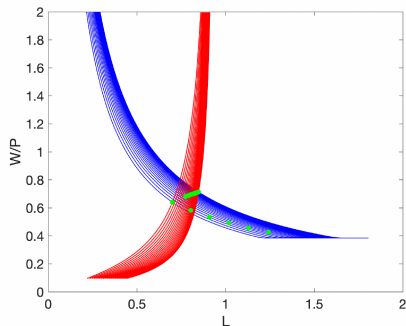


Table 1: Summary Statistics: Firm-Level Dataset.

	N	Mean	Std. dev.	10th	90th
Liabilities/Assets _{<i>i</i>,1918–1919}	798	0.47	0.18	0.23	0.69
$\Delta_{1919–1924}$ Liabilities/Assets _{<i>i</i>}	656	-0.25	0.20	-0.47	0.00
Debt/Assets _{<i>i</i>,1918–1919}	800	0.37	0.18	0.13	0.59
ln(Assets) _{<i>i</i>,1918–1919}	800	15.01	1.26	13.48	16.77
Fixed Assets/Assets _{<i>i</i>,1918–1919}	800	0.36	0.24	0.08	0.74
ROA _{<i>i</i>,1918–1919}	744	0.10	0.06	0.03	0.18
EBIT margin _{<i>i</i>,1918–1919}	758	0.36	0.22	0.10	0.68
Employment _{<i>i</i>,1919}	240	4,151.54	9,307.68	400.00	9,959.50
$\Delta \ln(\text{Employment})_{i,t}$	1,358	3.50	21.90	-2.90	19.24

Notes: This table reports summary statistics for the firm-level dataset based on *Saling's Boersenjahrbuch*. Variables with “1918 – 1919” subscripts are averaged over 1918 and 1919. $\Delta \ln(\text{Employment})$ is the change in log firm employment (times 100) from 1919 through 1923.